

# Radar Emitter Sorting Based on Multi-Head ResGAT

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## ABSTRACT

Conventional graph neural networks (GNNs) fail to effectively capture high-order relationships among radar pulses, thereby compromising discrimination accuracy in precise signal sorting. Therefore, this paper proposes a radar emitter signal sorting method based on an enhanced graph attention network (GAT). The model combines a multi-head attention mechanism with a residual network structure, enabling dynamic weight allocation to graph nodes. This effectively captures the complex correlation patterns of radar signals across a multi-dimensional parameter space and thus enhances classification performance. In scenarios with scarcely available labels and complex signal features, the proposed method demonstrates stronger average accuracy and robustness when handling radar signal sorting tasks.

## 1 | Introduction

Radar emitter signal sorting plays a crucial role in electronic reconnaissance. In recent years, with the rapid advancement of artificial intelligence, neural networks have received significant attention for their ability to autonomously extract and learn different radar pulse features. This addresses the problem of classifying radar signals from various radiation sources. Wu [1] utilised a PA sequence to train a GRU for prediction, while Liu [2] applied an RNN to extract temporal patterns for pulse sequence classification and denoising. While these networks can extract temporal information, they fail to account for the frequency variations of radar pulse signals. Furthermore, as supervised learning algorithms, these models rely heavily on large quantities of labelled data samples, a dependency that demands significant computational resources.

Current methods generally focus on the basic characteristics of pulse features (such as RF, PW, DOA, etc.), but they often overlook the implicit relational features between pulses. This limitation can result in decreased classification accuracy, particularly when dealing with interleaved complex signals, where

the distribution of pulse features is often challenging to differentiate. Graph neural networks (GNNs) have shown great potential in effectively integrating pulse correlation information [3]. However, conventional GNNs often face challenges in capturing the high-order relationships among radar signals within complex multi-dimensional parameter spaces, leading to constrained sorting accuracy. Graph attention networks (GATs) have demonstrated significant advantages in capturing these high-order relationships due to their ability to dynamically assign attention weights to different radar signals based on their features and interactions. Furthermore, GNNs encounter the “over-smoothing” challenge [4]. To mitigate the over-smoothing issue and enhance radar signal sorting accuracy, this paper presents a multi-head attention-based residual graph attention network (ResGAT) model.

## 2 | Proposed Method

The proposed radar emitter signal sorting method is structured as shown in Figure 1 and is composed of three main components: adaptive weighted pulse sequence graph

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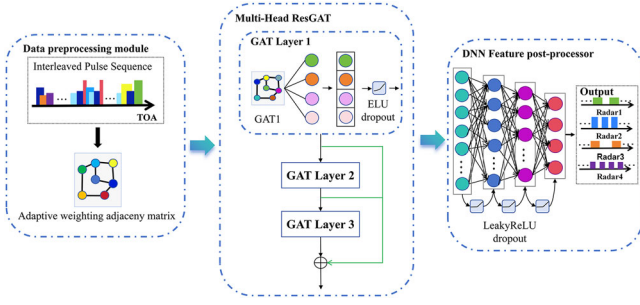


FIGURE 1 | Overview of proposed multi-head ResGAT architecture.

construction, graph-based representation learning and result prediction.

## 2.1 | Adaptive Weighted Pulse Sequence Graph Construction

As in our previous method [5], the adaptive distance weighting strategy is used to dynamically adjust the number of neighbours according to the local density of nodes so as to capture the local structure of data points more accurately.

## 2.2 | Structure of Multi-Head ResGAT

The model consists of multiple GAT layers, with residual connections linking each layer. The input to the first GAT layer is the node features  $\mathbf{h}_i^{(0)}$ , where the feature dimension of each node is (RF, PW, DOA). The node embeddings are computed through the GAT layers.

$$\mathbf{h}_i^{(1)} = \sigma \left( \sum_{j \in \mathcal{N}(i)} \alpha_{ij}^{(1)} \cdot \mathbf{W}^{(1)} \mathbf{h}_j^{(0)} \right) \quad (1)$$

where  $\alpha_{ij}^{(1)}$  represents the attention weights of the first multi-head attention layer, and  $\mathbf{W}^{(1)}$  is the weight matrix of the first layer. Then, the activation function used is exponential linear unit (ELU), and dropout is applied to reduce overfitting.

$$\mathbf{h}_i^{(1)} = \text{dropout}(\text{ELU}(\mathbf{h}_i^{(1)}), p = \text{feat\_dropout}) \quad (2)$$

The second and third layers follow the same principle. The outputs from the first and second layers are accumulated with the output of the third layer to form the final feature representation.

$$\mathbf{h}_i^{(3)} = \mathbf{h}_i^{(3)} + \mathbf{h}_i^{(1)} + \mathbf{h}_i^{(2)} \quad (3)$$

## 2.3 | DNN Feature Post-Processor

The DNN layer is stacked with multiple linear transformation layers, each of which uses a linear transformation, that is, a transformation of a feature through matrix multiplication.

$$\mathbf{y}^{(l)} = \mathbf{W}^{(l)} \mathbf{h}_i^{(3)} + \mathbf{b}^{(l)} \quad (4)$$

where,  $\mathbf{W}^{(l)}$  is the weight matrix of the  $l$ -th layer, which controls the influence of each input feature on the output.  $\mathbf{b}^{(l)}$  is the bias

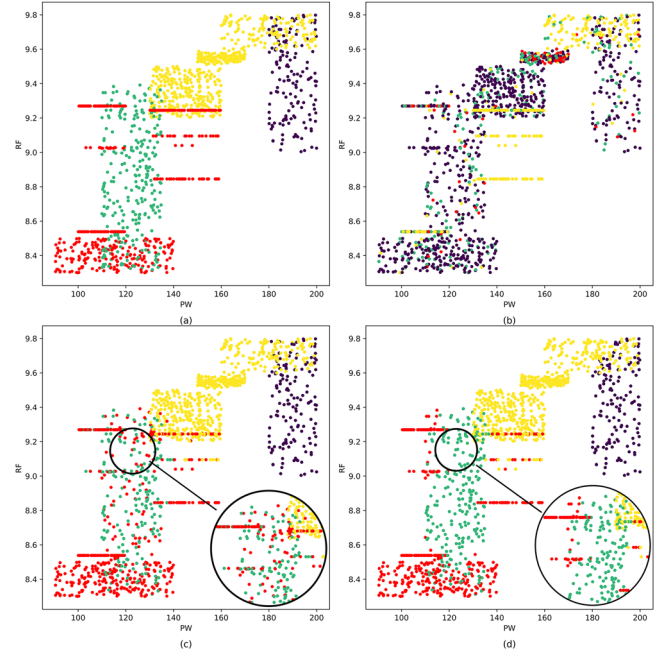


FIGURE 2 | Comparison of RESS results: (a) Original, (b) CNet, (c) ResGCN and (d) Multi-Head ResGAT(ours). The correspondence among radars and colours: red-radar1, green-radar2, yellow-radar3, purple-radar4.

term, allowing the model to adjust the output of each layer.  $\mathbf{h}_i^{(3)}$  is the output feature of the GAT. After the linear transformation at each layer, the LeakyReLU activation function is applied, followed by dropout to reduce overfitting.

$$\mathbf{y}_i = \left( \text{Linear} \left( \text{LeakyReLU} \left( \text{dropout} \left( \text{Linear} \left( \mathbf{h}_i^{(3)}, \mathbf{y} \right) \right) \right) \right) \right), x_i \quad (5)$$

After a series of fully connected layers, the output is normalised through the log\_softmax function at the output layer, allowing the output values to be interpreted as the probability distribution over the classes, resulting in the final classification output.

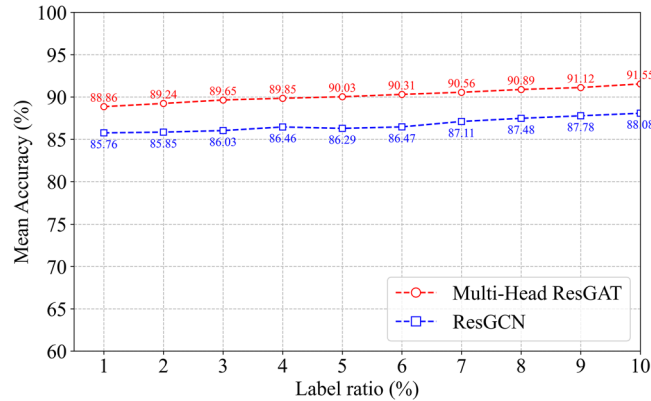
## 3 | Experimental Results and Analysis

Due to the lack of publicly available radar classification datasets, the simulated dataset setup described in reference [3] was adopted to validate the effectiveness of the proposed method. Four multifunctional radars were simulated, each configured with one to three different operating modes. DOA, PW, and RF were used as the primary features for each radar pulse, with distinct parameter values for each operating mode. To realistically simulate radar radiation source emission signals, overlap was introduced between different radars under the same parameters. Moreover, each parameter type was configured to use different modulation schemes, such as interleaving, hopping, jitter, and stagger. The visualised data and the results from different algorithms for this data are shown in Figure 2.

Experimental results in Table 1 demonstrate that the proposed method achieves the highest deinterleaving accuracy. Both the ADPC and CNet methods cannot fully separate the signals from the four radars, with accuracies of 43.20% and 54.03%,

**TABLE 1** | Performance comparison.

|          | Multi-Head<br>ResGAT | ResGCN-<br>RSS [1] | ADPC<br>[6] | CNet<br>[7] |
|----------|----------------------|--------------------|-------------|-------------|
| Radar1   | 99.65%               | 93.76%             | 48.67%      | 0           |
| Radar2   | 88.91%               | 84.89%             | 87.52%      | 32.46%      |
| Radar3   | 84.86%               | 74.12%             | 0           | 47.27%      |
| Radar4   | 94.98%               | 96.11%             | 22.22%      | 96.30%      |
| Accuracy | 91.55%               | 88.08%             | 43.20%      | 54.03%      |

**FIGURE 3** | Sorting accuracy comparison with different data labelled rates.

respectively. This is because both methods are based on peak density clustering, and when the radar signal parameters have a high degree of overlap, their classification approach, which relies on density similarity and data distribution characteristics, is not suitable in such cases. In contrast, our method and ResGCN-RSS successfully achieve precise separation of all four radars, underscoring the advantages of graph-based approaches for radar signal deinterleaving. At a 10% labelling rate, the overall deinterleaving accuracy reaches 91.66%, marking a 3.47% improvement over ResGCN-RSS. This enhancement is attributed to our method's flexible attention mechanism and the multi-head architecture's strong expressive power in handling heterogeneous data. These features boost the model's ability to capture complex multidimensional correlations between radar pulses and dynamically adjust aggregation ranges, effectively mitigating over-smoothing effects.

To evaluate the method's effectiveness under data-limited scenarios, comparative experiments with ResGCN-RSS across different label rates are presented in Figure 3. Our method consistently outperforms ResGCN-RSS, achieving an accuracy of 88.86% at a 1% label rate—surpassing ResGCN-RSS's peak performance (85.76%) even at a 10% label rate. This 3.10% improvement under extreme label scarcity highlights our framework's superiority in low-label scenarios.

## 4 | Conclusion

Conventional GNNs face challenges in capturing high-order correlations between radar pulses. This paper proposes an enhanced GNN-based radar emitter signal sorting method. By integrating

a multi-head attention mechanism with a residual network structure, the model effectively captures the complex interdependencies of radar signals across multi-dimensional parameter spaces, thereby improving classification performance. Simulation results indicate that the proposed method achieves superior average accuracy and robustness in scenarios with scarce labels and complex signal features.

## Author Contributions

**Liangang Qi:** conceptualisation, methodology, validation, writing – review and editing. **Hongzhuo Chen:** software, investigation, writing – original draft preparation and formal analysis. **Qiang Guo:** supervision, project administration and investigation. **Mykola Kaliuzhnyi:** writing – reviewing and editing.

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## Conflicts of Interest

The authors declare no conflicts of interest.

## Data Availability Statement

The data that support the findings of this study are available from the corresponding author upon reasonable request.

## References

1. J. Wu, L. Teng, and Q. Guo, "Radar Signal Sorting Based on GRU Neural Network," in Proc. IEEE 5th Inf. Technol., Netw., Electron. Automat. Control Conf., Xi'an, China, 2021, 711–714.
2. M. Liu and P. S. Yu, "Classification, Denoising and Deinterleaving of Pulse Streams With Recurrent Neural Networks," *IEEE Transactions on Aerospace and Electronic Systems* 55, no. 4 (August 2019): 1624–1639.
3. P. Lang, X. Fu, J. Dong, H. Yang, and J. Yang, "A Novel Radar Signals Sorting Method via Residual Graph Convolutional Network," *IEEE Signal Processing Letters* 30 (2023): 753–757.
4. D. Chen, et al., "Measuring and Relieving the Over-smoothing Problem for Graph Neural Networks From the Topological View," *Proceedings of the AAAI Conference on Artificial Intelligence* 34 no. 04 (2020).
5. L. Qi, H. Chen, Q. Guo, S. Huang, and M. Kaliuzhnyi, "GLS: A Hybrid Deep Learning Model for Radar Emitter Signal Sorting," *Digital Signal Processing* 161 (2025): 105117.
6. P. Lang, X. Fu, Z. Cui, C. Feng, and J. Chang, "Subspace Decomposition Based Adaptive Density Peak Clustering for Radar Signals Sorting," *IEEE Signal Processing Letters* 29 (2021): 424–428.
7. K. Chi, J. Shen, Y. Li, Y. Li, and S. Wang, "Multi-Function Radar Signal Sorting Based on Complex Network," *IEEE Signal Processing Letters* 28 (2020): 91–95.